Atlantic Sea Scallop Research Track Assessment - Community Engagement Meeting
December 18, 2024
Waypoint Event Center, New Bedford, MA and Webinar

MEETING SUMMARY

OVERVIEW

The Atlantic Sea Scallop Research Track Working Group[1] (Working Group) held a hybrid community engagement session on Wednesday, December 18, 2024 from 10:00 a.m. to 12:00 p.m. in New Bedford, MA and via webinar. The New England Fishery Management Council (Council) and NOAA Fisheries' Northeast Fisheries Science Center (NEFSC) jointly hosted the session. This meeting provided an opportunity for community members, including members of the scallop fishery, researchers, and the public, to engage in discussions about the current state of the scallop fishery and ongoing research efforts. Approximately 85 individuals attended, comprising Scallop Research Track Working Group members, Council staff, fishermen, industry representatives, and other community members.

Dr. Dvora Hart, lead stock assessment scientist for scallops, and Dr. Pat Sullivan, Chair of the Research Track Working Group, presented an overview of the research track assessment. They outlined the methodology for incorporating environmental and biological data into stock assessment models, including advancements in scientific techniques aimed at better understanding recruitment patterns, environmental influences, and predator impacts on scallops. The presentations were followed by a structured discussion that encouraged community members to share their observations and feedback.

Meeting materials are available at: https://www.nefmc.org/calendar/dec-18-2024-atlantic-sea-scallop-research-track-assessment-community-engagement-meeting

DISCUSSION QUESTIONS

The Working Group prepared a series of questions for community members to help guide discussion:

1. Have you observed changes in abundance in the following areas? Have you noticed any changes in the condition (shell thickness, size of abductor, disease) or mortality events in these areas?
    a. Inshore versus offshore areas? (Shallow vs. deeper areas). Have you been fishing in deeper water in recent years?
    b. Mid-Atlantic?
        i. Inshore? Southern? Long Island?
    c. Georges Bank?
    d. Gulf of Maine?
2. Have you noticed any changes in the timing of better meat yields versus the past? And if so:
    a. What do you think is causing this?
    b. Where are you observing these differences?
    c. Are there environmental factors that you think could contribute to this?
3. Similarly, have you noticed any changes to when scallops are spawning or appear to have spawned? If so, where did you observe this?
4. Which predators on scallops do you think are most important? Have you observed changes in the types, quantity, or location of these predators?
5. How has management influenced your decisions on when and where to fish? What changes to your fishing practices have you implemented over the past few years (e.g., in the last 5 years)? For example, the FT LA fleet has been allocated 24 DAS or fewer since 2018 and LA and IFQ allocations have declined each year since 2018. Have you changed where you sail from and/or

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where you land your scallops? Do you always target areas with higher catch rates, or are there other considerations?

6. Nematodes and shell blister disease have been reported in scallops in the Mid-Atlantic since the last assessment. Would you say that the prevalence is increasing or decreasing? How much discarding is happening in the cutting box due to meat quality issues, and has this changed over time? Are there other meat quality or disease issues of concern?
7. Is there information about the ecology or dynamics of this fishery that you think is important and/or that fisheries scientists may overlook or fail to understand? What are the most important research needs of the fishery?
8. Thoughts on performance of projection model and trying to predict scallops in an area at the end of its lifecycle. How can we improve? What is the rate of decline when conditions change? Year over year, how has our management worked?

QUESTIONS ON THE PRESENTATION

Dr. Hart and Dr. Sullivan fielded questions about the presentation. Members of the community raised several questions about the presentation, particularly concerning the availability and use of environmental data. One participant inquired about the availability of data that may show warming temperatures, to which Dr. Hart responded that temperature and other environmental factors are measured annually and made publicly accessible. She noted that recent years have shown unusual stratification of the thermocline in the Mid-Atlantic, with cooler bottom temperatures in 2024 despite warmer overall trends.

Questions about phytoplankton sampling were also addressed, with Dr. Hart explaining that this kind of data is collected using various methods, including satellite imagery and physical sampling. Unrelated to the focus of this outreach meeting, an attendee questioned the rationale behind the Council's suggestion to add four additional days-at-sea (DAS), asking if this implied an increase in scallop abundance. It was clarified that the decision was multifaceted and not necessarily linked to an immediate rise in scallop numbers. Instead, the measure considered recruitment trends and management goals.

A participant raised concerns about the models used for assessments, asking how much they rely on scientific versus experiential data. Dr. Hart answered that the models are primarily science-based, though there is some room for judgment in interpreting data. She acknowledged the importance of incorporating fishermen's observations and supported the idea of smaller, informal meetings to facilitate communication between scientists and fishermen. This sentiment was echoed by another attendee, who expressed frustration about the perceived lack of responsiveness to fishermen's input.

Dr. Sullivan and Dr. Hart reiterated that the New England Fishery Management Council (NEFMC) makes management decisions, and the research track focuses solely on the scientific basis for these policies. They encouraged participants to continue providing input, whether through future meetings or direct communication with the research team.

COMMUNITY DISCUSSION

Prior to the community discussion, Mayor Jon Mitchell of New Bedford welcomed attendees, voicing his support for the meeting and future consideration of access to the Northern Edge. Dr. Hart and Dr. Sullivan then reviewed the questions for the community, and discussion continued question by question.

1. *Have you observed changes in abundance in the following areas? Have you noticed any changes in the condition (shell thickness, size of abductor, disease) or mortality events in these areas?*

   a. *Inshore versus offshore areas? (Shallow vs. deeper areas). Have you been fishing in deeper water in recent years?*
   b. *Mid-Atlantic?*
      i. *Inshore? Southern? Long Island?*
   c. *Georges Bank?*
   d. *Gulf of Maine?*

Fishermen and industry representatives shared observations regarding changes in scallop abundance and condition across different regions. Many expressed significant concerns about heavy fishing pressure in areas reopened for harvest in recent years. They reported that these access areas often experience episodes of pulse fishing and/or heavy fishing pressure, leading to depletion of scallop beds in rotational areas. Several participants added that, in their view, management policies focusing fishing effort in concentrated areas result in diminished stock and habitat health, with insufficient attention given to the survival of smaller or juvenile scallops. They argued that open-bottom fishing allows for more natural recovery, as fishermen are more likely to leave areas with smaller scallops untouched. Attendees questioned the effectiveness of closed areas, especially when intense fishing is permitted as soon as these regions reopen. Some argued that such practices contradict the goals of sustainability and stock recovery, further exacerbating declines in abundance. Concerns were also raised about low survival rates among juvenile scallops in reopened areas.

In Area II, fishermen reported a notable decline in larger scallops. Another participant noted that while meats at 38-39 fathoms were good this season, scallops in deeper areas in Area II had poor yield, and the productive shallower region is now being heavily fished. Reports from the Mid-Atlantic Open Areas and New York Bight reflected similar trends, with one fisherman stating that fishing off Long Island has not been productive for several years, and fishermen had not observed seasonal improvements in meat yields as they had expected.

There was a comment on the need for real-time data to better understand current fishing conditions and scallop yields. It was proposed that auction data could be used as a source of real-time information, allowing for greater transparency and more informed decision-making for those involved in the fishery.

2. *Have you noticed any changes in the timing of better meat yields versus the past? And if so:*
   a. *What do you think is causing this?*
   b. *Where are you observing these differences?*
   c. *Are there environmental factors that you think could contribute to this?*

Fishermen reported noticeable changes in the timing of higher meat yields compared to previous years. A commercial fisherman with over five decades of experience noted that scallops during their prime harvesting periods no longer exhibit the same growth patterns seen in the past. He highlighted that despite hauling large numbers of scallops aboard, the actual meat yield has been disappointing, with smaller and less robust meats becoming increasingly common. Another attendee shared observations of significant week-to-week variability in meat counts, particularly in Closed Area II. This unpredictability in yield was noted as a departure from more consistent patterns observed historically.

Fishermen in Great South Channel and Northern and Southern Flank noted that meats did not improve at any point during the season, suggesting a disruption in the usual seasonal patterns. Participants attributed these issues to environmental factors such as warming waters, as well as management practices forcing fishing during less optimal months.

3. *Similarly, have you noticed any changes to when scallops are spawning or appear to have spawned? If so, where did you observe this?*

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Fishermen reported significant changes in scallop spawning patterns compared to previous years, with many noting that healthy gonads have become increasingly rare. Several participants stated that they had not observed robust spawning activity during the current season, which they found unusual and concerning. One individual pointed out that gonads, which typically indicate spawning readiness, have appeared smaller and less developed, further supporting the observation of disrupted spawning cycles.

4. *Which predators on scallops do you think are most important? Have you observed changes in the types, quantity, or location of these predators?*

Fishermen highlighted several predators as significant threats to scallop populations, with sea stars being the most frequently mentioned. Participants reported an increase in sea star populations, particularly in Georges Bank, and noted that these predators have been observed preying heavily on juvenile scallops. Concerns were raised about the growing prevalence of Astropecten sea stars, which thrive in warmer waters and have expanded their range from deeper areas into shallower waters as temperatures have risen.

Suggestions were made to explore predator control measures, such as grinding sea stars on vessels to limit their impact. However, some participants cautioned against such practices, emphasizing the need to ensure that only harmful species are targeted. One attendee pointed out that improperly managing predator control could disrupt the ecosystem, as not all sea stars prey on scallops, and some may even help control other harmful species. Another participant noted that cutting sea stars into pieces could unintentionally increase their population, as certain species can regenerate from fragments.

One participant commented that current management practices direct concentrated fishing effort into access areas, leading to depletion of the local resource and habitat. They voiced concerns that fishermen are essentially being instructed to deplete specific areas, resulting in negative future impacts for the fishery. They highlighted poor fishing practices, such as deck loading, and argued that management policies forcing many vessels into small areas are as much to blame as individual practices. They called for more research into the recovery times for fished areas and the long-term impacts of such concentrated efforts on scallop populations and their predators.

There was a proposal to leverage existing reporting systems to better monitor predator impacts. All fishermen are already required to complete Vessel Trip Reports (VTRs) after each trip, so the creation of a user-friendly policy to allow them to report predator sightings and other observations was suggested.

5. *How has management influenced your decisions on when and where to fish? What changes to your fishing practices have you implemented over the past few years (e.g., in the last 5 years)? For example, the FT LA fleet has been allocated 24 DAS or fewer since 2018 and LA and IFQ allocations have declined each year since 2018. Have you changed where you sail from and/or where you land your scallops? Do you always target areas with higher catch rates, or are there other considerations?*

Fishermen expressed frustration that management policies often force them to fish during suboptimal times for meat quality. One participant highlighted that current regulations allow fishing in April, when scallop meat yields tend to be smaller, instead of allowing them to focus efforts in June and July, which are traditionally better months for meat yield. This timing, they argued, undermines both the quality of the product and the sustainability of the stock.

Challenges with fixed gear were also highlighted. In Area II, one participant noted that lobster gear in the eastern portion of the fishing areas had been left in place, creating access issues for scallop vessels.

6. *Nematodes and shell blister disease have been reported in scallops in the Mid-Atlantic since the last assessment. Would you say that the prevalence is increasing or decreasing? How much discarding is happening in the cutting box due to meat quality issues, and has this changed over time? Are there other meat quality or disease issues of concern?*

Fishermen noted that nematodes and shell blister disease remain persistent issues in the Mid-Atlantic, with some suggesting their prevalence has increased in recent years. One participant observed that nematodes are becoming more common in the Mid-Atlantic. Another described significant discards due to poor meat quality, particularly in areas like New York Bight, where scallops with diseased or poor-quality meats tend to be discarded while vessels continue to fish the access area until they achieve their trip limit. This practice, they argued, leads to waste and further stock depletion.

The discussion also touched on research gaps, particularly regarding the impact of wind farms on scallop health and distribution. Fishermen asked whether studies are being conducted to assess how wind farm development affects water column dynamics and larval transport.

Concerns about the observer program were also raised, with some fishermen questioning the accuracy and reliability of observer-collected data. One participant suggested that a greater focus on data collected directly by fishermen during trips could improve the reliability of assessments. Chad Keith (NEFSC Fisheries Monitoring and Operations) encouraged participants to report specific instances of observer inaccuracies to address these concerns.

7. *Is there information about the ecology or dynamics of this fishery that you think is important and/or that fisheries scientists may overlook or fail to understand? What are the most important research needs of the fishery?*

There was no discussion on this question

8. *Thoughts on performance of projection model and trying to predict scallops in an area at the end of its lifecycle. How can we improve? What is the rate of decline when conditions change? Year over year, how has our management worked?*

Fishermen expressed mixed opinions on the performance of the projection models used to predict scallop populations, particularly in areas nearing the end of their lifecycle. Several participants criticized the models for failing to accurately capture the rapid decline in scallop abundance once environmental conditions shift or areas are heavily fished. One attendee noted that management often allocates too many boats to small, concentrated areas, leading to swift depletion and limited recovery opportunities for the resources there.

Suggestions for improvement included incorporating more real-time data from the field to adjust projections dynamically as conditions change. Fishermen emphasized the importance of studying recovery rates for heavily fished areas to better understand long-term sustainability.

With no additional discussion, Dr. Hart and Dr. Sullivan thanked the participants for their time and input. The meeting adjourned at 12:30PM

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